

CSE211-Discrete Mathematics
Fall 2023
Midterm Exam
November 24, 2023
Duration: 75 mins

Student no: Answer
Full Name: Key
Instructor: Dr. Tülay AYYILDIZ

This is an exam in English, answers in other languages will not be accepted.

Grading table:

Questions:	1	2	3	4	5	Total
Max point:	20	20	20	20	20	100
Points						

QUESTIONS

1. (20 points) (10+10p) Prove or disprove $\neg(p \wedge \neg q) \vee q \equiv p \Rightarrow q$

(a) by using a truth table

(b) without using a truth table.

2. (20 points) (6+6+8p) Let $A = \{1, 2, \{3\}, a\}$, $B = \{\{1, 2\}, a\}$ and $C = \{1, \{2, 3\}\}$. What are the following sets:

(a) $A \cup B$

(b) $C \cap B$

(c) $A \times B$

3. (20 points) (7+7+6p) Consider $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = x^2 - 2$ on integers. Is $f(x)$ one-to-one, onto, bijection?

4. (20 points) What is the following sum

$$\sum_{i=10}^{100} i(i-1) = ?$$

5. (20 points) We define the sequence of numbers as

$$a_n = \begin{cases} 1 & \text{if } 0 \leq n \leq 3, \\ a_{n-1} + a_{n-2} + a_{n-3} + a_{n-4} & \text{if } n \geq 4 \end{cases}$$

Prove or disprove that $a_n \equiv 1 \pmod{3}$ for all $n \geq 0$.

SOLUTIONS

(a)

#1.

P	q	$\neg q$	$P \wedge \neg q$	$\neg(P \wedge \neg q)$	$\neg(P \wedge \neg q) \vee q$	$P \Rightarrow q$
T	T	F	F	T	T	T
T	F	T	T	F	F	F
F	T	F	F	T	T	T
F	F	T	F	T	T	T

$$\begin{aligned} (b) \quad \neg(P \wedge \neg q) \vee q &\equiv \neg P \vee q \vee q \\ &\equiv \neg P \vee q \\ &\equiv P \Rightarrow q \end{aligned}$$

$$\#2 (a) A \cup B = \{1, 2, \{3\}, a, \{1, 2\}\}$$

$$(b) C \cap B = \emptyset$$

$$(c) A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

$$= \{(1, \{1, 2\}), (1, a), (2, \{1, 2\}), (2, a), (\{3\}, \{1, 2\}), (\{3\}, a), (a, \{1, 2\}), (a, a)\}$$

#3. Given: $f: \mathbb{Z} \rightarrow \mathbb{Z}$
 $x \mapsto x^2 - 2$

⊗ One-to-one? Definition: For all $x_1, x_2 \in \mathbb{Z}$,
 If $f(x_1) = f(x_2)$ then $x_1 = x_2$.

1st approach: (Proof by contradiction)

Assume $f(x_1) = f(x_2)$ for all $x_1, x_2 \in \mathbb{Z}$

then $x_1^2 - 2 = x_2^2 - 2$

$x_1^2 = x_2^2$

$\Rightarrow x_1 = -x_2$ and $x_1 = x_2$

contradiction

f is not one-to-one

2nd approach: (Counter example)

For $x_1 = 1$ and $x_2 = -1$,

$f(x_1) = f(x_2)$ but $x_1 \neq x_2$. Thus f is not one-to-one.

⊗ onto? Definition: For all $x \in \mathbb{Z}$, $\exists y \in \mathbb{Z}$ such that $y = f(x)$.

1st approach: (Direct proof)

For all $x \in \mathbb{Z}$, assume that $\exists y$ s.t. $y = x^2 - 2$

then $\sqrt{y+2} = x$. $\sqrt{y+2}$ is not an integer for some values of y . Therefore f is not onto.

2nd approach: (Counter example)

For $0 \in \mathbb{Z}$, $\nexists y$ such that $0 = x^2 - 2 \Rightarrow x = \sqrt{2} \notin \mathbb{Z}$.

⊗ f is not bijective since it is neither one-to-one nor onto.

#4. $\sum_{i=10}^{100} i(i-1) = \sum_{i=10}^{100} i^2 - i = \sum_{i=10}^{100} i^2 - \sum_{i=10}^{100} i$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$= \left[\sum_{i=1}^{100} i^2 - \sum_{i=1}^9 i^2 \right] - \left[\sum_{i=1}^{100} i - \sum_{i=1}^9 i \right]$$

$$= \left(\frac{100 \cdot 101 \cdot 201}{6} - \frac{9 \cdot 10 \cdot 19}{6} \right) - \left(\frac{100 \cdot 101}{2} - \frac{9 \cdot 10}{2} \right)$$

$$= \frac{10}{6} (10 \cdot 101 \cdot 201 - 9 \cdot 19) - 5005 = 333\,060$$

6: First let see the structure of the given sequence:

#5 $a_0=1$
 $a_1=1$
 $a_2=1$
 $a_3=1$
 $a_4 = a_3 + a_2 + a_1 + a_0 = 4 \equiv 1 \pmod{3}$
 $a_5 = a_4 + a_3 + a_2 + a_1 \equiv 1 + 1 + 1 + 1 \equiv 1 \pmod{3}$
 $a_6 = a_5 + a_4 + a_3 + a_2 \equiv 1 + 1 + 1 + 1 \equiv 1 \pmod{3}$

Note:

(We solved this kind of induction with 2 steps in the base and the assumption here, we do it for 4 proper steps)

Proof by induction:

Base: For $n=0,1,2,3$

$a_0=a_1=a_2=a_3=1 \equiv 1 \pmod{3}$ by the definition of the seq.

Assume that for $n=k, k-1, k-2$ $a_n \equiv 1 \pmod{3}$

Now we want to show if $n=k+1$ $a_n \stackrel{?}{\equiv} 1 \pmod{3}$

For $n=k+1$

$$a_{k+1} = a_k + a_{k-1} + a_{k-2} + a_{k-3}$$

By our assumption

$$\begin{aligned} a_k &\equiv 1 \pmod{3} \Rightarrow \exists m_0 \in \mathbb{Z} \text{ s.t. } a_k = 3m_0 + 1 \\ a_{k-1} &\equiv 1 \pmod{3} \Rightarrow \exists m_1 \in \mathbb{Z} \text{ s.t. } a_{k-1} = 3m_1 + 1 \\ a_{k-2} &\equiv 1 \pmod{3} \Rightarrow \exists m_2 \in \mathbb{Z} \text{ s.t. } a_{k-2} = 3m_2 + 1 \\ a_{k-3} &\equiv 1 \pmod{3} \Rightarrow \exists m_3 \in \mathbb{Z} \text{ s.t. } a_{k-3} = 3m_3 + 1 \end{aligned}$$

$$\begin{aligned} a_{k+1} &= (3m_0+1) + (3m_1+1) + (3m_2+1) + (3m_3+1) \\ &= 3(m_0+m_1+m_2+m_3) + 4 \\ &\equiv 1 \pmod{3} \end{aligned}$$

Thus for all $n \geq 4$ $a_n \equiv 1 \pmod{3}$.

Note: There are other methods to prove this. I accepted your solutions if they were correct.